

工业中的有限元方法

——作业二

一、 作业任务

利用二次基函数的有限元方法求解：

$$\begin{cases} -\frac{d^2u}{dx^2} + u = 0, 0 < x < 1 \\ u(0) = 0 \text{ 和 } u(1) = 1 \end{cases}$$

二、 基本原理

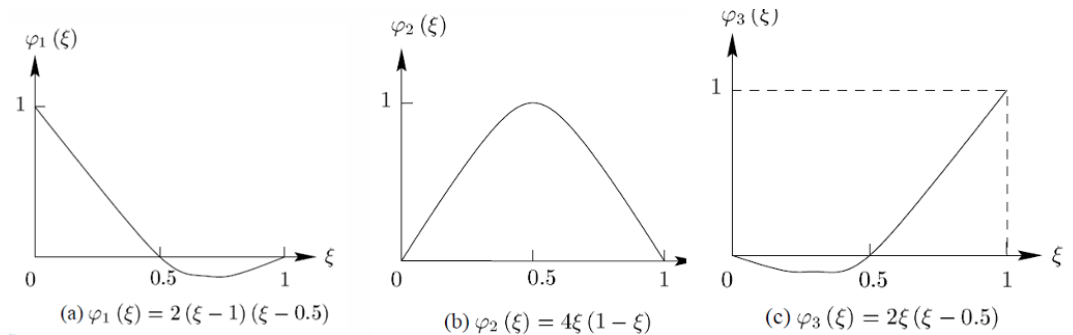


图 1 二次基函数

利用二次基函数求解刚度矩阵，则此时由三对角矩阵转化为五对角矩阵。

求解可得如下：

$$a(\varphi^2, \varphi^0) = \int_0^1 (4x-1)(4x-3)dx = \frac{1}{3}$$

$$a(\varphi^2, \varphi^1) = 4 \int_0^1 (4x-1)(1-2x)dx = -\frac{8}{3}$$

$$a(\varphi^2, \varphi^2) = \int_0^1 (4x-1)^2 dx = \frac{7}{3}$$

$$a(\varphi^1, \varphi^0) = 4 \int_0^1 (4x-3)(1-2x)dx = -\frac{8}{3}$$

$$a(\varphi^1, \varphi^1) = 16 \int_0^1 (1-2x)^2 dx = \frac{16}{3}$$

$$a(\varphi^0, \varphi^0) = \int_0^1 (4x-3)^2 dx = \frac{7}{3}$$

三、实验结果

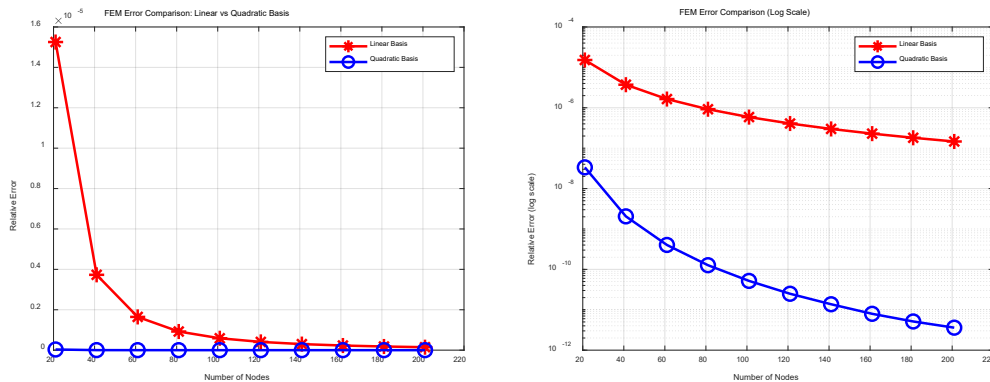


图 2 一次基函数与二次基函数误差对比

如图 2 所示，为了直观分析，绘制了对数坐标。总体来看，图像直观表明在该一维 FEM 问题中，二次基函数精度优于一次基函数，增加节点数可有效降低误差。

四、实验代码

函数 fem1dquadratic: 实现二次基函数有限元求解

```
function un = fem1dquadratic(u0, u1, n, xn)
% Assemble the stiffness matrix A for quadratic elements
A = zeros(n, n);
% Reference matrices K_ref and M_ref for quadratic elements
K_ref = [7/6, -4/3, 1/6;
         -4/3, 8/3, -4/3;
         1/6, -4/3, 7/6];
M_ref = [4/15, 2/15, -1/15;
         2/15, 16/15, 2/15;
         -1/15, 2/15, 4/15];
num_elements = (n - 1)/2; % Number of quadratic elements
```

```

for k = 1:num_elements
    left = 2*k -1;
    mid = 2*k;
    right = 2*k +1;
    h = xn(right) - xn(left);
    % Compute local matrix contribution
    local_A = (2/h) * K_ref + (h/2) * M_ref;
    % Assemble into global matrix
    A(left, left) = A(left, left) + local_A(1,1);
    A(left, mid) = A(left, mid) + local_A(1,2);
    A(left, right) = A(left, right) + local_A(1,3);
    A(mid, left) = A(mid, left) + local_A(2,1);
    A(mid, mid) = A(mid, mid) + local_A(2,2);
    A(mid, right) = A(mid, right) + local_A(2,3);
    A(right, left) = A(right, left) + local_A(3,1);
    A(right, mid) = A(right, mid) + local_A(3,2);
    A(right, right) = A(right, right) + local_A(3,3);
end

% Apply boundary conditions
internal_nodes = 2:n-1;
% Construct system matrix and RHS
A_system = A(internal_nodes, internal_nodes);
b_system = -A(internal_nodes, 1)*u0 - A(internal_nodes, n)*u1;
% Solve the linear system
un_system = A_system \ b_system;
% Combine the solution
un = zeros(n,1);
un(1) = u0;
un(end) = u1;
un(internal_nodes) = un_system;
end

```

主函数:

```
% 设置公共参数
a1 = 0; a2 = 1; % 区间 [0, 1]
u0 = 0; u1 = 1; % 边界条件
m = 10; % 细化次数

% 初始化误差数组
err_linear = zeros(m, 1);
err_quadratic = zeros(m, 1);

% 线性基函数的误差计算
for k = 1:m
    n_linear = k*10 + 1; % 线性节点数
    xn_linear = linspace(a1, a2, 2*(n_linear - 1) + 1).';
    un_linear = femldlinear(u0, u1, 2*(n_linear - 1) + 1, xn_linear); % 假设返回内部节点解
    % 精确解在内部节点
    uex_linear = exp(1)/(exp(2)-1) * (exp(xn_linear(2:end-1)) - exp(-xn_linear(2:end-1)));
    err_linear(k) = norm(un_linear - uex_linear) / norm(uex_linear);
end

% 二次基函数的误差计算
for k = 1:m
    n_linear = k*10 + 1; % 对应的线性节点数
    n_quadratic = 2*(n_linear - 1) + 1; % 二次节点数
    xn_quadratic = linspace(a1, a2, n_quadratic).';
    un_quadratic = femldquadratic(u0, u1, n_quadratic, xn_quadratic); % 返回所有节点解
    % 精确解在内部节点 (排除边界)
    uex_quadratic = exp(1)/(exp(2)-1) * (exp(xn_quadratic(2:end-1)) - exp(-xn_quadratic(2:end-1)));
    err_quadratic(k) = norm(un_quadratic(2:end-1) - uex_quadratic) / norm(uex_quadratic);
end

% 绘制对比图
figure;
plot((1:m)*20 + 1, err_linear, '-*r', 'DisplayName', 'Linear Basis', 'LineWidth', 2, 'MarkerSize', 11);
hold on;
plot((1:m)*20 + 1, err_quadratic, '-ob', 'DisplayName', 'Quadratic Basis', 'LineWidth', 2, 'MarkerSize', 11);
hold off;
title('FEM Error Comparison: Linear vs Quadratic Basis');
xlabel('Number of Nodes');
ylabel('Relative Error');
legend('show');
grid on;

% 使用对数坐标以更清晰显示收敛阶
figure;
semilogy((1:m)*20 + 1, err_linear, '-*r', 'DisplayName', 'Linear Basis', 'LineWidth', 2, 'MarkerSize', 11);
hold on;
semilogy((1:m)*20 + 1, err_quadratic, '-ob', 'DisplayName', 'Quadratic Basis', 'LineWidth', 2, 'MarkerSize', 11);
hold off;
title('FEM Error Comparison (Log Scale)');
xlabel('Number of Nodes');
ylabel('Relative Error (log scale)');
legend('show');
grid on;
```